ASML : Operation and characterization of the Reticle Handling Robotic Arm

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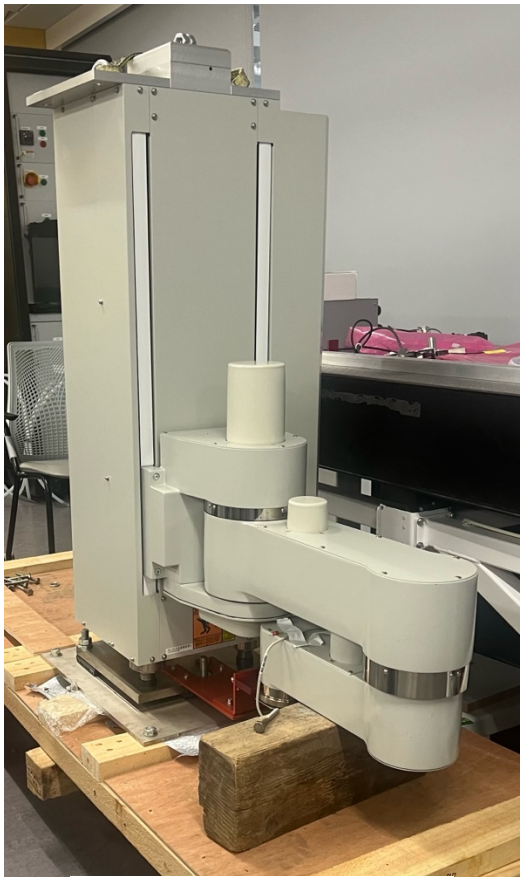

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1. Executive Summary

1.1 Project Description

This project focuses on the characterization and analysis of the ASML SR7150 Reticle Handling Robotic Arm, a multi-joint industrial manipulator used for precision handling applications. While the broader project scope includes system integration, safety planning, and control infrastructure development, this MEng project specifically concentrates on evaluating the mechanical, kinematic, and dynamic limitations of the robotic arm. A physics based analytical model was developed to quantify the robot's workspace, mass properties, stability margins, and load induced torque requirements under representative operating conditions. The analysis was conducted to support the safe mechanical integration and operation of the system on the supporting stand.

1.2 Purpose of the characterization of the SR7150 Reticle Handling Robotic Arm

The purpose of the characterization was to establish performance limits for the robotic arm. In particular, the analysis was aimed to identify the workspace boundaries, center of mass behavior, static tipping margins, and dynamic stability during motion. Understanding these limitations are critical for preventing unsafe operating configurations, and guiding control strategies, and informing mechanical integration decision when the robot is deployed on a supporting stand.

1.3 Summary of outcomes and recommendations

The static stability analysis using reaction force based tipping criterion showed that the system remains stable under gravitation loading for the evaluated range of the stand masses and base offset. The dynamic screening analysis indicated that the highest inertial load occur near extended arm configuration representing the worst case operating condition. Based on these results, the robotic arm can be safely integrated onto the stand.

2. Introduction

2.1 Motivation

The primary motivation for this project is to enable the safe and effective use of the ASML SR7150 Reticle Handling Robotic Arm within the Mechatronics Learning Studio. Due to the robot's significant mass, reach, and cost, the system cannot be mounted on a table without first verifying that the supporting structure would not introduce a tipping or stability risk. An uncontrolled tip-over event would pose a serious safety hazard and results in substantial equipment damage, making stability verification and critical prerequisite for use of the system.

2.2 Project Overview

This MEng project focuses on the analytical characterization of the ASML SR7150 Reticle Handling Robotic Arm with an emphasis on workspace, mass distribution, and stability limitations. The work involved developing a simplified kinematic model of the arm, computing its reachable workspace under joint constraints, and estimating center of mass across representative configurations.

Static stability was evaluated using gravitational loading and reaction force based tipping criteria, while dynamic stability was through Jacobian based velocity estimation and conservation stopping assumption. All analysis was implemented in MATLAB to enable parametric studies and identify the worst-case operating conditions. The results of this work provide a quantitative bound on safe operation and serve as the foundation for future mechanical integration.

3. Robot Characterization

3.1 Mechanical Overview

The ASML SR7150 Reticle Handling Robotic Arm is modeled as a planar, two-link serial manipulator mounted to a fixed base. The arm consists of two revolute joints, referred to as TH1 and TH2, which enables positioning of the end effector.

A simplified kinematic model was developed to capture the essential geometric behavior of the arm while enabling tractable analytical analysis. Link lengths, joint types, and joint limits were defined based on available system specification and physical inspection. Table 1 summarizes the primary mechanical and kinematic parameters used in the model.

Parameter	Symbol	Value	Unit
TH1 (Link 1 length)	L_1	.400	m
TH2 (Link 2 length)	L_2	.300	m
TH1 joint limits	θ_1	-90 to 90	degrees
TH2 joint limits	θ_2	-170 to 170	degrees

Table 1: Mechanical and kinematic Parameters

3.2 Workspace & Reach Analysis

The reachable workspace of the manipulator was evaluated using a forward kinematic model for a planar two-link arm. The Cartesian position of the end effector was expressed as a function of the joint angles and link lengths according to the following relations:

$$\boldsymbol{x} = L_1 \sin\theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

Equation 1: Motion in the x-direction

$$\boldsymbol{y} = L_1 \cos\theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

Equation 2: Motion in the y-direction

By sweeping the joint angles across their allowable ranges, the full reachable workspace of the end effector was computed and visualized to identify the kinematic limits and extreme configurations, as shown in Figure 1.

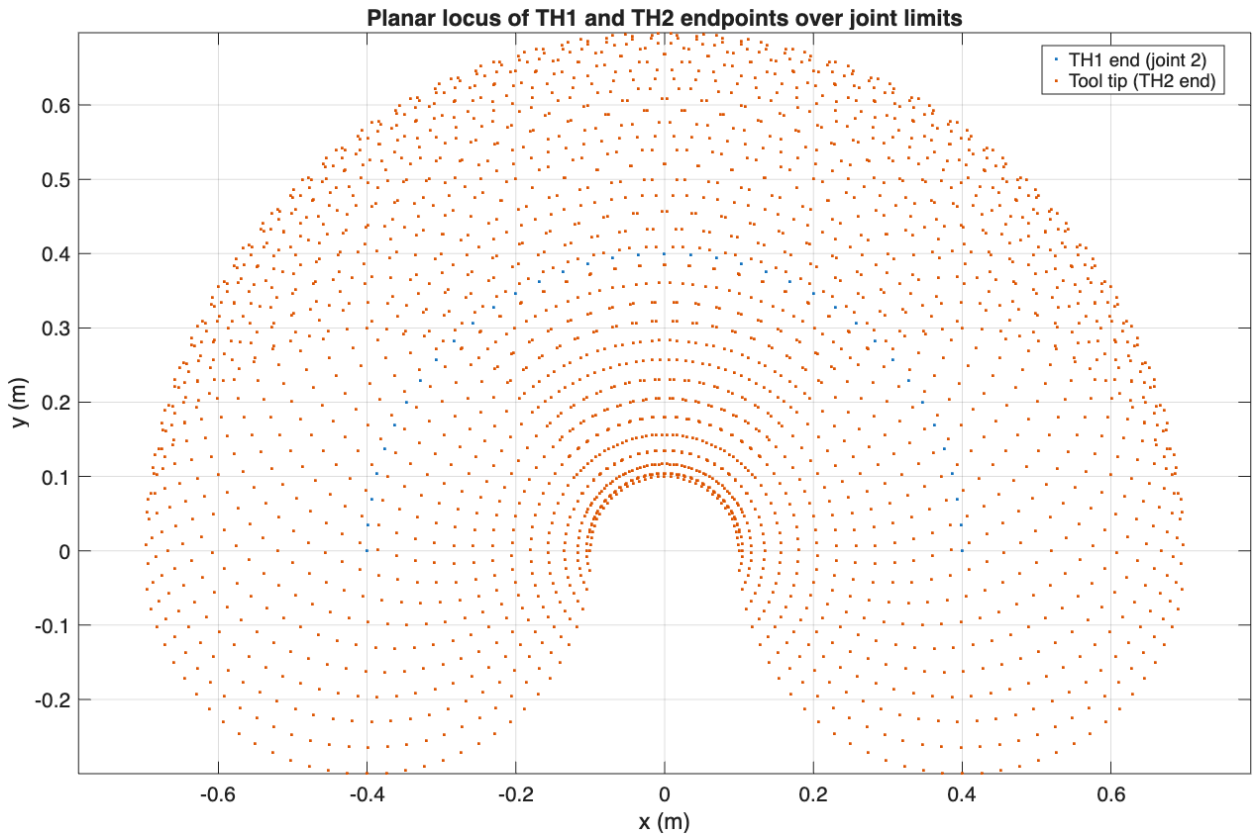

Figure 1: Planar locus of TH1 and TH1 endpoints over joint limits

3.3 Mass Properties & COM

To evaluate stability and base loading, the mass properties of the robotic arm were approximating lumped masses located at representative center of mass positions along each link. The vertical column links TH1 and TH2, and the tool/payload were each modeled as discrete mass elements. Table 2 summarizes the mass properties as assumed center of mass location used in the analysis.

Component	**Mass (kg)**	**Symbol for Mass**	**COM Location from Base (m)**	**Symbol for COM**
Vertical column	140	m_{box}	0.000	x_{box}
TH1 link	15	m_1	0.200	x_1
TH2 link	10	m_2	0.550	x_2
Tool + payload	6	x_T	0.700	x_T
Total Mass	171	m_r	N/A	N/A

Table 2: COM of components of Manipulator

3.4 Motion Constraints Used for Dynamic Analysis

While this project does not focus on detailed control system implementation, key control related parameter where incorporated into the analysis to assess dynamic effect during motion. Maximum joint velocities and conversative stopping times were used to estimate worst case end-effector accelerations. Table 3 summarizes the control parameters used in the dynamic stability analysis.

Parameter	Value	Units
Max TH1 joint speed	2.22	rad/s
Max TH2 joint speed	2.79	rad/s
Assumed stopping time	0.15	s

Table 3: Control Parameters for the Robotic Arms

These parameters were used in conjunction with eh manipulator Jacobian to estimate end-effectors velocities and inertial load during rapid deceleration events. The resulting acceleration estimates form the basis for the dynamic stability screen presented in section 4.4.

4. Stability, Dynamics, and Torque Calculations

4.1 Inputs & Assumptions

The stability analyses presented in this chapter are based on the simplified mechanical and kinematic model introduced in section 3. All calculations were performed using rigid body approximation and are intended to provide conservative estimates of the worst-case loading and stability behavior.

The joint angles were defined relative to a fixed base reference frame with TH1 measured directly from the base and the TH2 defined relative to the orientation of the first link. Due to the physical orientation of the TH2 joint of the robotic arm, an angle transformation was applied to maintain a consistent reference frame for the kinematic and stability calculations. TH2 joint

angle is expressed as $\theta_2 = 180° - \theta_{2,spec}$, where $\theta_{2,spec}$ corresponds to the joint angle TH2 arm is angled at. The transform ensures that joint angle corresponds the consistent physical rotation and direction within the planar mode. All joint angles were converted to radians prior to numerical computations.

The robot max torque was not included in the stability analysis and the evaluation focused solely on the loading and inertial effected associated with the worst case kinematic motion.

4.2 Static Stability – Robot + Stand (Venation Method)

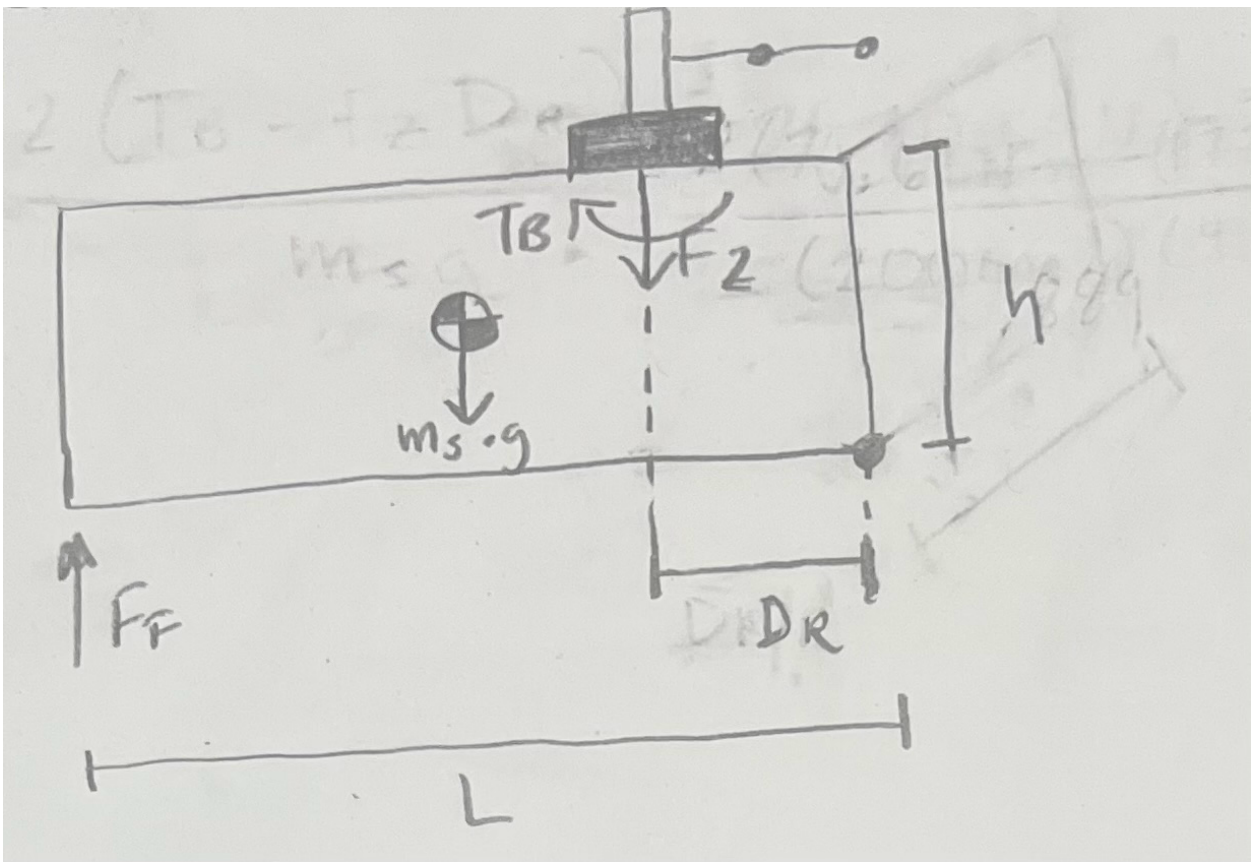

Figure 2: Free body Diagram of the Robot with the stand

Static stability of the robot stand was evaluated using a reaction force-based tipping criterion adapted from the Vention robot stand stability methodology. A free body diagram of the robot mounted on the supporting stand is shown in Figure 2.

First, the overall center of mass of the robotic arm was calculated using a lumped mass approximation:

$$x_{r,COM} = \frac{m_{box} * x_{box} + m_1 * x_1 + m_2 * x_2 + m_T * x_T}{m_r}$$

Using the mass properties defined in section 3.3, the robot center of mass was found to be:

$$x_{r,COM} = 0.0743\ m$$

The total gravitation force acting on the robot was then computed as:

$$F_Z = m_r * g = 1677.51\ N$$

The resulting overturning moment applied at the robot base due to gravity was:

$$T_B = m_r * g * x_{r,COM} = 124.59\ N \cdot m$$

Next, the reaction force at the front edge of the stand, F_f, was calculated by summing the moment about the rear of the stand:

$$F_f = \frac{F_Z * D_R + m_s g s - T_B}{L}$$

where Dr is 0.6096 m is the distance of the robot base to the front edge of the stand, m_s is the stand mass, and L is the stands length. According to the stability criterion, the system is considered statically stable if $F_f > 0$.

To evaluate stability across a range of possible support structure, the stand mass was varied from 400 kg to 1500 kg using MATLAB. The critical base offset corresponding to the tipping condition ($F_f = 0$) was computed as:

$$D_{R,crit} = \frac{T_B - m_s * g * \frac{L}{2}}{F_Z}$$

For cases where if $D_{R,crit}$> 0, a safety factor was defined as:

$$SF = \frac{D_R}{D_{R,crit}}$$

A value of SF > 1 indicated a stable configuration, while a SF < 0 indicated a tipping risk. However, if $D_{R,crit} < 0$, the tipping point lies in front of the stand footprint, indicating that the system is stable for any meaning value of D_R.

The resulting stability metrics for the range of the stand masses considered are summarized in Table 2. For all evaluated stand masses, the calculated $D_{R,crit}$ values were negative, which indicated the robot stand system is statically stable under gravitational loading for the specified base offset.

m_table	Dr_crit	Fr_actual	SF_desc
(kg)	(m)	(N)	
400	-2.778	2330.3	No tipping for any Dr>=0
500	-3.491	2820.8	No tipping for any Dr>=0
600	-4.204	3311.3	No tipping for any Dr>=0
700	-4.917	3801.8	No tipping for any Dr>=0
800	-5.63	4292.3	No tipping for any Dr>=0
900	-6.343	4782.8	No tipping for any Dr>=0
1000	-7.056	5273.3	No tipping for any Dr>=0
1100	-7.769	5763.8	No tipping for any Dr>=0
1200	-8.482	6254.3	No tipping for any Dr>=0
1300	-9.195	6744.8	No tipping for any Dr>=0
1400	-9.907	7235.3	No tipping for any Dr>=0

Table 2: Static Stability Results for Varying Stand Mass

4.3 Dynamic Stability During Motion

Dynamic stability during robot motion was evaluated using a Jacobian based velocity and acceleration analysis. The forward kinematic relationship defined in section 3.2 were differentiated to obtain the manipulator Jacobian:

$$J = \begin{bmatrix} \frac{\partial x}{\partial x} & \frac{\partial x}{\partial y} \\ \frac{\partial y}{\partial x} & \frac{\partial y}{\partial y} \end{bmatrix} = \begin{bmatrix} L_1 \cos\theta_1 + L_2 \cos(\theta_1 + \theta_2) & L_2 \cos(\theta_1 + \theta_2) \\ -L_1 \sin\theta_1 - L_2 \sin(\theta_1 + \theta_2) & -L_2 \sin(\theta_1 + \theta_2) \end{bmatrix}$$

The end effector velocity was then computed as:

$$V_{tool} = J * \dot{\theta}$$

Where the joint velocity vector is defined as:

$$\dot{\theta} = \begin{bmatrix} 2.22 \\ 2.79 \end{bmatrix} \frac{rad}{s}$$

To estimate the worst case internal loading during the rapid deceleration, the tool accelerations component were approximated using a stopping time T_{stop}:

$$a_x = \frac{V_{tool}(1)}{T_{stop}} \; and \; a_y = \frac{V_{tool}(2)}{T_{stop}}$$

These accelerations were evaluated across the full joint space, and the resulting acceleration magnitudes were visualized over the workspace. The maximum acceleration values occurred near extended arm configuration.

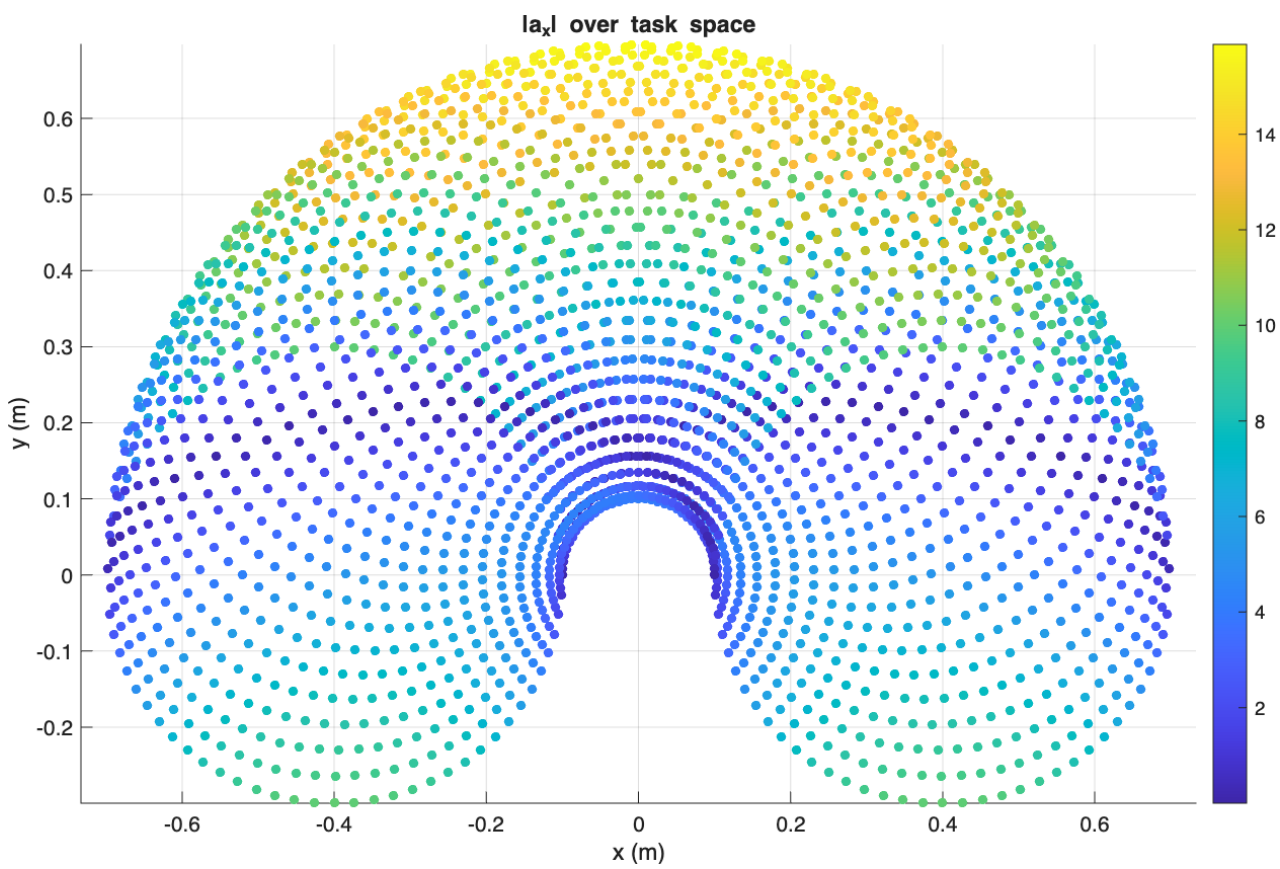

Figure 3: |a_x| over task space

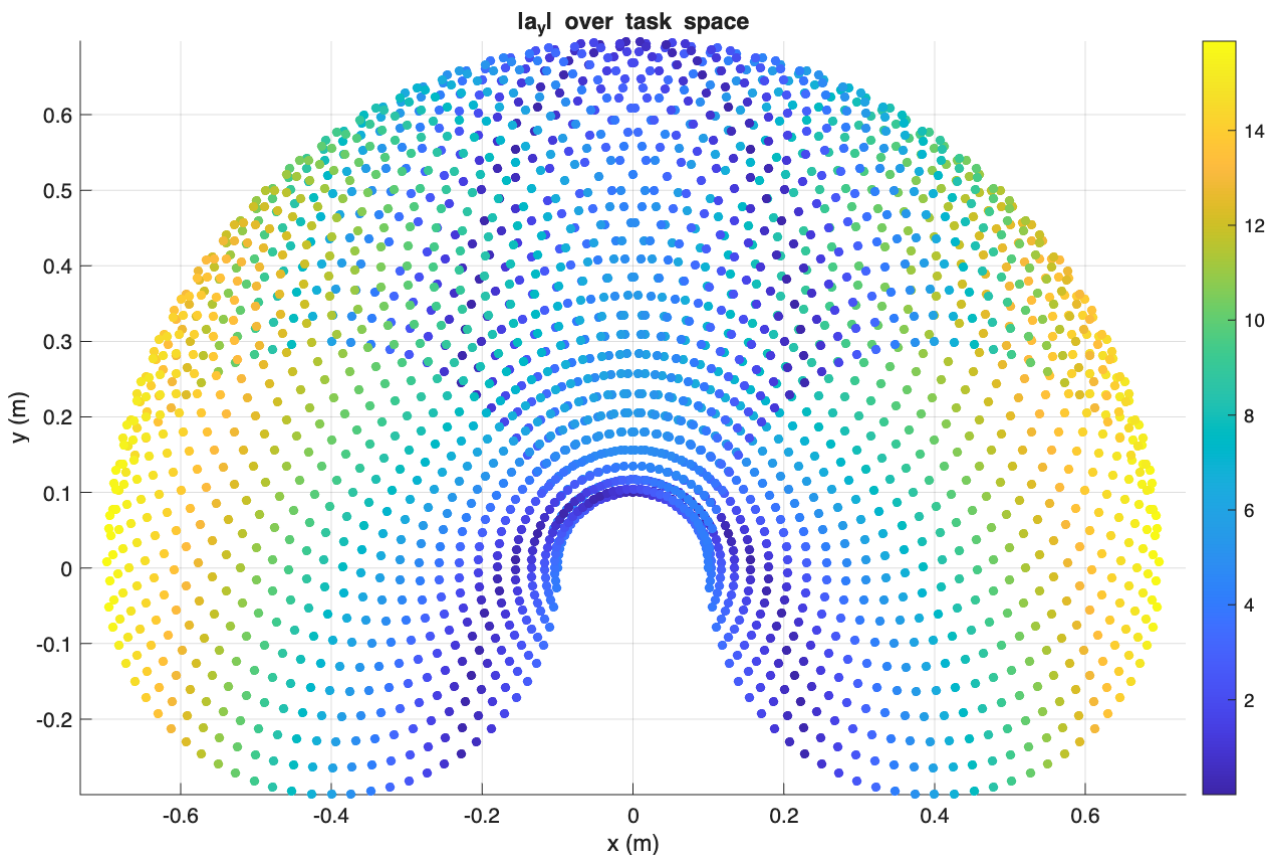

Figure 4: |a_y| over task space

These results indicate that the highest dynamic loads occur near extended arm configurations, which present worse-case operating condition for stability considerations

5. Appendices

5.1 Full MATLAB script

```
%ASML: Operation and characterization of the Reticle Handling Robotic Arm


% Initialize parameters for the robotic arm
TH1_length = .400; % Length of the TH1 arm
TH2_length = .300; % Length of the TH2 arm
totalLength = TH1_length + TH2_length; % Calculate the total length of the
robotic arm
m_box = 140; % mass of the vertical column (has z-axis mechanism, motors
structure (units: kg)
m_TH1 = 15; % mass of TH1 (unit: kg)
m_TH2 = 10; % mass of TH2 (unit: kg)
m_t = 6; % mass of the tool + payload

% Initialize parameters for the table stand
L = 2.4384; % Length of the table (unit: m)
```

```
W = .762; % width of the table (unit: m)
H = .889; % Height of the table (unit: m)
m_table = 2000; % mass of the table (unit: kg)

% Center of mass (COM) positions for the robotic arm
x1 = .200; %  COM of the TH1 (units: m)
x2 = .550; %  COM of the TH2 (units: m)
xt = .700; %COM of the tool + payload (units: m)

% Calculate the total mass of the robotic arm system
totalMass = m_box + m_TH1 + m_TH2 + m_t;


% Calculate the center of mass (COM) for the entire robotic arm system
COM_x = (m_box * 0 + m_TH1 * x1 + m_TH2 * x2 + m_t * xt) / totalMass;

%% === ROBOT-ONLY: BASE TORQUE STATIC ===

% Calculate the gravitational force acting on the robotic arm system
g = 9.81; % Acceleration due to gravity (m/s^2)
Fz = totalMass * g; % Total weight of the robotic arm system

% Calcualte the torque on base
torque_base = Fz*COM_x; %overturning moment from the robot about the base
% Robot base distance from FRONT edge (this is Vention's Dr)
Dr_actual = .6096; % Distance from the edge of the table

% Range of table masses to study
m_table_vec = 400:100:1500;   % kg

fprintf('m_table   Dr_crit    Fr_actual       SF_desc\n');
fprintf('  (kg)      (m)        (N)\n');

for mt = m_table_vec

    % 1) Critical Dr where rear reaction Fr = 0 (tipping limit)
    Dr_crit = (torque_base - mt*g*(L/2)) / Fz;

    % 2) Rear reaction at your actual Dr
    Fr_actual = (mt*g*(L/2) + Fz*Dr_actual - torque_base) / L;

    % 3) Safety factor interpretation + description
    if Dr_crit > 0
        SF = Dr_actual / Dr_crit;
        if Fr_actual <= 0
            SF_desc = sprintf('UNSTABLE (Fr<=0, SF=%.2f)', SF);
        elseif SF > 1
            SF_desc = sprintf('Stable, SF=%.2f', SF);
        elseif abs(SF - 1) < 1e-3
            SF_desc = 'At tipping (SF~1)';
        else
            SF_desc = sprintf('UNSTABLE (SF=%.2f<1)', SF);
        end
    else
        % Dr_crit < 0: tipping point is in front of table (Dr < 0)
```

```
        SF_desc = 'No tipping for any Dr >= 0';
    end

    fprintf('%6.0f   %7.3f   %10.1f   %s\n', ...
        mt, Dr_crit, Fr_actual, SF_desc);

end

%% === TH1 & TH2 WORKSPACE (with TH2 stop near straight) ===
L1 = TH1_length;
L2 = TH2_length;

% TH1 joint limits from spec
theta1_deg = -90:5:90;

% TH2 joint limits from spec
theta2_deg = -170:5:170;

[TH1_deg, TH2_deg] = meshgrid(theta1_deg, theta2_deg);

% Angle convention:
% a1 = TH1 angle
a1 = deg2rad(TH1_deg);
a2 = deg2rad(180-TH2_deg);

% TH1 end (joint 2) position
x1_end = L1 .* sin(a1);
y1_end = L1 .* cos(a1);

% TH2 end (tool) position - standard 2-link planar model
x_tool = L1 .* sin(a1) + L2 .* sin(a1 + a2);
y_tool = L1 .* cos(a1) + L2 .* cos(a1 + a2);

figure;
plot(x1_end(:), y1_end(:), '.', 'MarkerSize', 4); hold on;
plot(x_tool(:), y_tool(:), '.', 'MarkerSize', 4);
axis equal;
xlabel('x (m)');
ylabel('y (m)');
legend('TH1 end (joint 2)', 'Tool tip (TH2 end)', 'Location','Best');
title('Planar locus of TH1 and TH2 endpoints over joint limits');
grid on;


%% === 4.3 Dynamic Stability over workspace via Jacobian ===

L1 = TH1_length;
L2 = TH2_length;

% Joint limits (you already have these)
theta1_deg = -90:5:90;
theta2_deg = -170:5:170;
[TH1_deg, TH2_deg] = meshgrid(theta1_deg, theta2_deg);

a1 = deg2rad(TH1_deg);           % TH1 in rad
```

```
a2 = deg2rad(180 - TH2_deg);     % your angle convention

% Joint max speeds (given)
dq = [2.22; 2.79];   % [rad/s]  TH1, TH2

t_stop = 0.15;       % assumed hard-stop time [s]

%% === 4.3 Dynamic Stability over workspace via Jacobian ===

L1 = TH1_length;
L2 = TH2_length;

% Joint limits (you already have these)
theta1_deg = -90:5:90;
theta2_deg = -170:5:170;
[TH1_deg, TH2_deg] = meshgrid(theta1_deg, theta2_deg);

a1 = deg2rad(TH1_deg);           % TH1 in rad
a2 = deg2rad(180 - TH2_deg);     % your angle convention

% Joint max speeds (given)
dq = [2.22; 2.79];   % [rad/s]  TH1, TH2

t_stop = 0.15;       % assumed hard-stop time [s]

%% === 4.3 Dynamic Stability over workspace via Jacobian ===

L1 = TH1_length;
L2 = TH2_length;

% Joint limits (you already have these)
theta1_deg = -90:5:90;
theta2_deg = -170:5:170;
[TH1_deg, TH2_deg] = meshgrid(theta1_deg, theta2_deg);

a1 = deg2rad(TH1_deg);           % TH1 in rad
a2 = deg2rad(180 - TH2_deg);     % your angle convention

% Joint max speeds (given)
dq = [2.22; 2.79];   % [rad/s]  TH1, TH2

t_stop = 0.15;       % assumed hard-stop time [s]

% Matrices to store positions, velocities, accelerations
x_mat  = zeros(size(a1));
y_mat  = zeros(size(a1));
vx_mat = zeros(size(a1));
vy_mat = zeros(size(a1));
a_x_mat = zeros(size(a1));
a_y_mat = zeros(size(a1));


for i = 1:size(a1,1)
    for j = 1:size(a1,2)
        th1 = a1(i,j);
```

```
        th2 = a2(i,j);

        % Forward kinematics (same convention as earlier)
        x = L1*sin(th1) + L2*sin(th1+th2);
        y = L1*cos(th1) + L2*cos(th1+th2);

        % Jacobian based on your x,y defs
        J = [ L1*cos(th1) + L2*cos(th1+th2),   L2*cos(th1+th2);
             -L1*sin(th1) - L2*sin(th1+th2),  -L2*sin(th1+th2) ];

        v_tool = J * dq;      % [vx; vy] in m/s

        % Estimated deceleration to stop in t_stop
        ax = v_tool(1) / t_stop;
        ay = v_tool(2) / t_stop;

        % Store everything
        x_mat(i,j)   = x;
        y_mat(i,j)   = y;
        vx_mat(i,j)  = v_tool(1);
        vy_mat(i,j)  = v_tool(2);
        a_x_mat(i,j) = ax;
        a_y_mat(i,j) = ay;
    end
end

% Worst-case accelerations over all joint combos
a_x_max = max(abs(a_x_mat(:)));
a_y_max = max(abs(a_y_mat(:)));

fprintf('\n=== Worst-case dynamic accelerations over workspace (t_stop = %.2f
s) ===\n', t_stop);
fprintf('max |a_x| = %.2f m/s^2\n', a_x_max);
fprintf('max |a_y| = %.2f m/s^2\n', a_y_max);



% Flatten everything into column vectors
theta1_col = TH1_deg(:);   % deg
theta2_col = TH2_deg(:);   % deg
x_col      = x_mat(:);
y_col      = y_mat(:);
vx_col     = vx_mat(:);
vy_col     = vy_mat(:);
ax_col     = a_x_mat(:);
ay_col     = a_y_mat(:);

% Build table
dynTable = table(theta1_col, theta2_col, ...
                 x_col, y_col, ...
                 vx_col, vy_col, ...
                 ax_col, ay_col, ...
    'VariableNames',
{'TH1_deg','TH2_deg','x_m','y_m','vx_mps','vy_mps','ax_mps2','ay_mps2'});
```

```
% Look at first few rows
disp(dynTable(1:10,:));


% Scatter of position colored by |ax|
figure;
scatter(dynTable.x_m, dynTable.y_m, 20, abs(dynTable.ax_mps2), 'filled');
colorbar;
xlabel('x (m)');
ylabel('y (m)');
title('|a_x| over task space');
axis equal;
grid on;

% Scatter of position colored by |ax|
figure;
scatter(dynTable.x_m, dynTable.y_m, 20, abs(dynTable.ay_mps2), 'filled');
colorbar;
xlabel('x (m)');
ylabel('y (m)');
title('|a_y| over task space');
axis equal;
grid on;
```